

Lomash Relia

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Focus Areas: Space Robotics, Deep Learning, Computer Vision, Edge Computing, Physical AI

EDUCATION

ABV-Indian Institute of Information Technology and Management <i>M.Tech. in Autonomous Systems and Machine Intelligence</i> CGPA (Sem 1): 9.71/10	Gwalior, India Aug. 2025 – Present
Charotar University of Science and Technology <i>B.Tech. in Computer Engineering</i> CGPA: 9.41/10	Anand, India Sep. 2021 – Jun. 2025

EXPERIENCE

Teaching Assistant <i>ABV-IIITM Gwalior</i>	Jan. 2026 – Present Gwalior, India
- Assisting in laboratory and tutorial sessions for Android Application Development for second year B.Tech. students. - Helping students strengthen problem-solving skills by focusing on error interpretation and business logic design.	
IEEE MTTTS Student Branch Secretary <i>ABV-IIITM Gwalior</i>	Dec. 2025 – Present India
MTP Student <i>ABV-IIITM Gwalior (Mentor/Supervisor - Dr. Somesh Kumar)</i>	Aug. 2025 – Present Gwalior, India
- Developing mathematical control systems for the SO-101 robotic arm using MuJoCo based simulation. - Exploring vision language action (VLA) models for robotic manipulation in space and planetary exploration tasks. - Working with Bambu Lab A1 and prototype/print robotic parts	
Research Intern <i>Space Applications Center, ISRO</i>	Jan. 2025 – Jun. 2025 Ahmedabad, India
- Developed prototype planetary rover with Raspberry Pi featuring monocular depth estimation, manual and automated control modes via bluetooth/hotspot PAN, and modular architecture, along with a stereo photogrammetry lunar simulation - Quantized and optimized deep learning models for edge deployment, achieving near real-time inference while minimizing performance degradation	
Research Collaborator <i>Indian Institute of Tropical Meteorology</i>	Aug. 2024 – Nov. 2024 Pune, India
- Engineered deep learning pipeline for 4-channel Tropical Cyclone Image-to-Intensity Regression - Analyzed multi-channel satellite imagery data and neural network architectures for intensity estimation	
Full Stack Engineer <i>Capitalize</i>	May 2024 – Jul. 2024 Remote
- Engineered Flutter applications with FastAPI backend integration and RESTful API development - Automated data collection pipelines using Selenium and rotating proxies achieving 99% success rate	
Flutter Development Intern <i>Toshal Infotech Pvt. Ltd.</i>	May 2023 – Jun. 2023 Surat, India
Worked on Flutter application development and gained hands-on experience with Flutter architecture.	

- Developed mobile application components using Bloc for state management and Dio for network communication.
- Explored integration of machine learning features within mobile applications.

PROJECTS

Eye-in-Hand IBVS for SO-101 Robot Arm | *MuJoCo, OpenCV, Python* Dec. 2025 – Jan. 2026

- Implemented layered eye-in-hand Image-Based Visual Servoing (IBVS) in MuJoCo for the open-source 5-DoF SO-101 robot arm using ArUco marker detection, radial multi-depth search, Chaumette's pixel-based control law with damped pseudo-inverse, marker-size depth estimation, and lever-arm compensation for camera offset
- Engineered robust pipeline with joint/velocity limits, minimum gripper height safety, automatic divergence recovery (restart on failure), real-time visualization with feature overlays, and JSON-based feature teaching tool, achieving reliable convergence to 30–40 px feature error

AI-Powered Planetary Rover (Pragyan-Inspired) | *PyTorch, OpenCV* Jan. 2025 – Jul. 2025

- Designed and deployed complete rover system running on Bookworm OS with onboard edge AI capabilities
- Implemented monocular metric depth estimation and real-time object detection while keeping system temperature capped at 65°C
- Developed dual control modes supporting autonomous navigation and manual control with remote operability via VNC

Deep Learning for Cyclone Intensity Estimation | *PyTorch, TensorFlow* Sep. 2024 – Oct. 2024

- Replicated and validated rotation-blended CNN methods from published research papers
- Benchmarked model performance against published results for 4-channel satellite imagery analysis

LLM-Based Knowledge Extraction System | *FastAPI, LangChain, RAG* April 2024

- Developed knowledge graph generation system using LLMs and LangChain for text-to-knowledge transformation
- Implemented Retrieval-Augmented Generation (RAG) pipeline for enhanced information retrieval

POC Mobile App for Brain Tumor Detection | *Flutter, TensorFlow Lite* May 2023 – Jun. 2023

- Integrated TensorFlow Lite brain tumor detection model into Flutter mobile application
- Optimized deep learning model for mobile deployment achieving real-time inference performance

TECHNICAL SKILLS

Languages: Python, C++, SQL, Dart

Deep Learning & AI: NLP, LLMs, VLMs, GANs, VLA, Computer Vision

Robotics: Inverse Kinematics, Edge AI, Control Systems, MuJoCo, ROS2

Frameworks: FastAPI, PyTorch, TensorFlow, LangChain, LlamaIndex, OpenCV, Flask, Flutter, Selenium

Tools: Linux, PostgreSQL, Git, Firebase, VNC

Hardware: Raspberry Pi, Servo Control, Sensor Integration

ACHIEVEMENTS

- Achieved Rank 52 in KEA PG CET 2025 (M.Tech) examination
- Secured All India Rank 4687 in GATE 2025 (Data Science & AI) (normalized score of 413)
- Published article in WGNE Blue Book (Nov. 2024) and research in Springer (Jul. 2024)
- Presented work at ICHCSC Conference (Jul. 2024) and PRL Venus Science Conference (Sep. 2024)
- Presented work at IEEE INGARSS 2025 Symposium (10–13 Dec. 2025)